MEDICAL PROFESSIONALS

In the 18th century health at sea increasingly became a subject of systematic inquiry and naval medicine was slowly professionalized. Theories of disease were still woefully inadequate—the fumes from smelly cheese were thought a serious health risk by the Royal Navy. But systematic studies by ships' surgeons such as James Lind, whose *Treatise on Scurvy* was published in 1753, encouraged action by captains and administrators to improve cleanliness and nutrition. The role of citrus fruit and fresh vegetables in preventing scurvy was known from at least the early 17th century, but organizing a general policy to eradicate the disease was a slow business. Still, by 1800 the British fleet blockading Brest could keep 24 ships at sea all through a summer without crippling death rates.



SURGEON'S INSTRUMENTS A typical naval surgeon's chest contained instruments for amputation, trephination, draining, dentistry, probing wounds, and bleeding, as well as copious bandages and cloth for tourniquets.

SUNRAY TREATMENT Submarine crews back from patrol take sunray treatment in the depot ship to compensate for their lack of exposure to daylight when on active duty. The goggles are to protect their eyes.



> *I shall here only observe that the result of all my experiments was that oranges and lemons were the most effectual remedies for this distemper at sea.*
>
> **James Lind,** *A Treatise of the Scurvy,* 1753

SICK BERTH Located far away from living quarters to prevent the spread of disease, the sick berth on an 18th-century sailing ship was constructed out of canvas screens. During battle the screens were taken down and patients moved to the orlop level for safety.



HOSPITAL SHIP Wounded US marines lie in their bunks in close quarters aboard the USS *Bolivar* which transferred casualties in the course of its service during World War II.



CUTTING THE DEATH RATE

The introduction of steamships was a boon. They were healthier above all because they cut voyage times. Food on board improved sharply through the 19th century, and water quality was ameliorated by storage in iron tanks. The general progress of medicine and provision of properly qualified medical staff reduced deaths from disease and wounds radically by the 20th century. The nature of wounds in battle altered, with the treatment of flash burns an unexpected problem for ships' medical staff in World War I. But as in the wider body of society, epidemic disease became a radically diminishing threat.



ONBOARD OPERATION Cooperation between navies means that an injured American sailor can be taken 300 miles (480 km) to a British aircraft carrier where surgeons perform damage control surgery on his fractured ankle in an onboard operating room.